

Introduction to Fault-Tolerant Quantum Circuits

AlgoLab Summer School — Lecture 1

Adam Bene Watts

Prelude: Why Build a Quantum Computer?

A Quantum Industry Snapshot

- **Over \$10B** in funding from 2012–2024.
 - **\$2B** recently announced by the US Government (CHIPS and Science Act).



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- **1B+** in industry revenue in 2025.

Sources: MIT Quantum Index Report 2025 (qir.mit.edu);

QED-C State of the Global Quantum Industry 2026 (quantumconsortium.org)



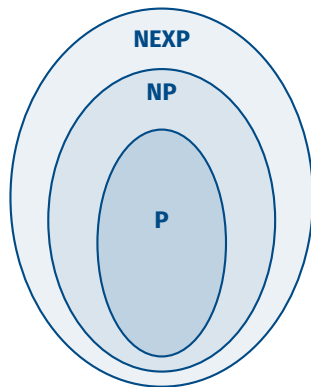
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Where It All Began: The Extended Church-Turing Thesis

- The **Church-Turing Thesis**: any function computable by a physical process can be computed by a Turing machine.

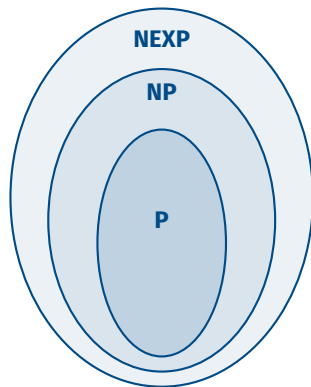
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- The **Church-Turing Thesis**: any function computable by a physical process can be computed by a Turing machine.
- The **Extended Church-Turing Thesis**: any function *efficiently* computed by a physical process can be *efficiently* computed by a Turing machine.



Where It All Began: The Extended Church-Turing Thesis

- The **Church-Turing Thesis**: any function computable by a physical process can be computed by a Turing machine.
- The **Extended Church-Turing Thesis**: any function *efficiently* computed by a physical process can be *efficiently* computed by a Turing machine.
- **Reasoning**: Any physical process can be efficiently simulated by a Turing machine. So if a physical process efficiently computes a function, the Turing machine can simulate that process, then read the result.



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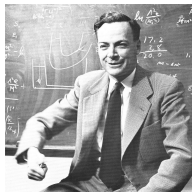
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“Nature isn’t classical, dammit, and if you want to make a simulation of nature, you’d better make it quantum mechanical.”

— Richard Feynman, 1981



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- A device which processes information using the laws of quantum mechanics *may not* be efficiently simulable by a Turing machine.
- Such a device *may* be able to efficiently solve problems which cannot be efficiently solved by any other classical physical process.

Part 1: Physical Qubits

One (Physical) Qubit States

- A (physical) **qubit** models the behavior of a **two-level** quantum system.



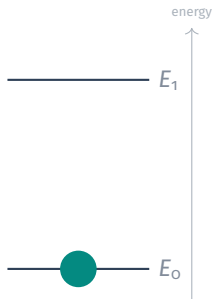
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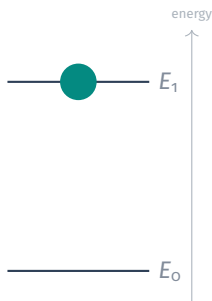
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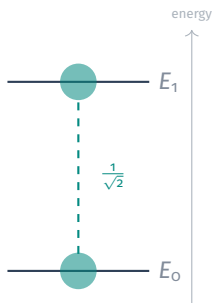
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 - $|\psi\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$: the system is in a **superposition** of the two states.



- A state of a qubit is a norm-one vector in some **Hilbert space**.
 - **Hilbert Space**: A vector space with inner product, complete in the induced norm.
 - For finite dimensions the unique choice (up to isomorphism) is $\mathbb{C}^n$ with the standard inner product.

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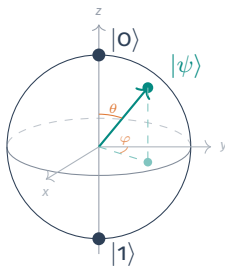
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Math Recap

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- A useful geometric representation is the **Bloch sphere**.



$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

One (Physical) Qubit Gates

- Physically, single-qubit gates evolve $|\psi\rangle$ according to the Schrödinger equation:

$$i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle \quad \implies \quad |\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle$$

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Question: Assume we can rotate around X and Z:

$$R_X(\theta) = e^{-i\theta X/2}, \quad R_Z(\theta) = e^{-i\theta Z/2}$$

Can we implement any single-qubit unitary? (And how?)

Compiling a Single Qubit Unitary

Q: Can we implement any single unitary with R_X and R_Z gates? **A:** Yes!

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- Since the Bloch sphere represents states up to global phase, we can write:

$$|0\rangle \mapsto e^{i\theta_1} |\psi_b\rangle, \quad |1\rangle \mapsto e^{-i\theta_1} |\psi_b^\perp\rangle$$

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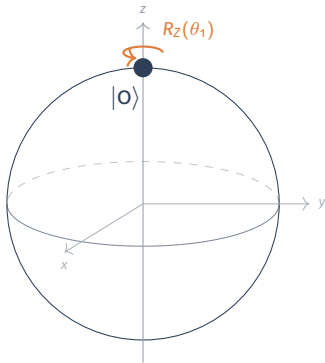
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- We can then decompose U as:

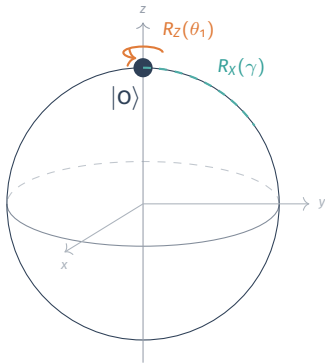
$$U = R_Z(\gamma) \cdot R_X(\beta) \cdot R_Z(\theta_1)$$

where $R_Z(\theta_1)$ contributes the phase, and $R_X(\beta) \cdot R_Z(\gamma)$ rotates to the correct point.

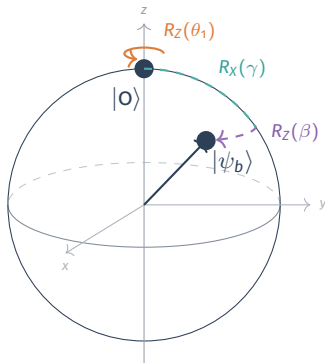
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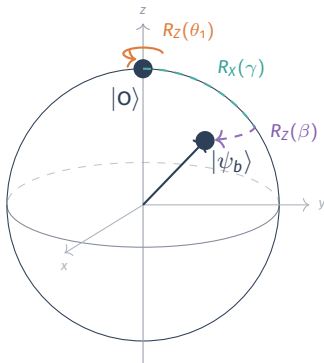


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Note: This also shows that every single-qubit unitary can be represented by a rotation of the Bloch sphere.

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Exercise(s):

1. (Easy) How many real numbers does it take to describe an n -qubit state? An n -qubit product state?
2. (Harder) How many bits to describe each to within error ε ? (In say, ℓ_2 norm?)

From your answers, explain why product states can be described classically while general states cannot.

Measuring Many Qubit States

- Recall:
 - A **projective valued measure** is a set of projectors $\{P_1, \dots, P_k\}$ summing to the identity.
 - Applying the measurement to $|\psi\rangle$ returns outcome i with probability $\|P_i |\psi\rangle\|^2$, leaving the system in state $P_i |\psi\rangle / \|P_i |\psi\rangle\|$.

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- But we can perform many others — and these need not be destructive! For example:

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- Applying $\{P_{\text{even}}, P_{\text{odd}}\}$ to $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ returns outcome P_{even} with probability 1 and leaves the state unchanged:

$$P_{\text{even}} |\Phi^+\rangle = |\Phi^+\rangle.$$

Many-Qubit Unitaries

- Say we can engineer a two-qubit interaction $H \propto ZZ$. Evolving for the correct time gives the **CZ gate**:

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Questions: Using only CZ and single-qubit unitaries, can we implement:

1. (Hard) Any unitary of the form CU for arbitrary U ?
2. (Very Hard) Any two-qubit unitary?
3. (Very Hard) Any multi-qubit unitary?

Many-Qubit Unitaries: Proof 1

Q: Can we implement any CU gate using CZ and single-qubit gates?

A: Yes!

Proof Sketch:

- Find matrices A, B, C such that $ABC = I$ and $AZBZC = U$.

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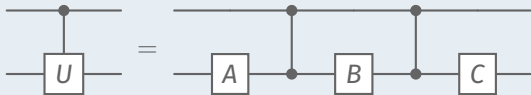
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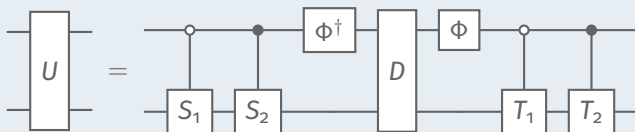


Many-Qubit Unitaries: Proof 2

Q: Can we implement any two-qubit unitary using CZ and single-qubit gates? **A:** Yes!

Proof Sketch:

- By the *cosine-sine decomposition*, any two-qubit unitary U can be written as:

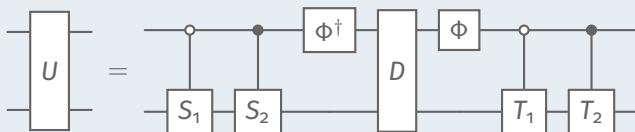


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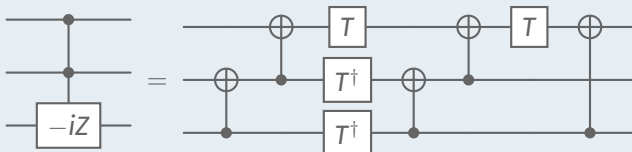
- Then apply the result from Proof 1.

Many-Qubit Unitaries: Proof 3

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Proof Sketch:

- First show we can implement multi-controlled Z gates using the identity:



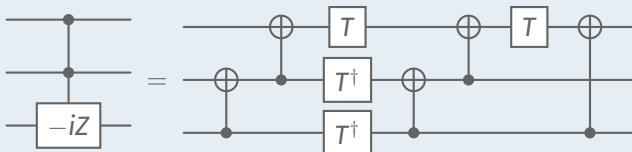
Identity from *Quantum Circuits of T-depth 1*, Peter Selinger, <https://arxiv.org/abs/1210.0974>

Many-Qubit Unitaries: Proof 3

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Proof Sketch:

- First show we can implement multi-controlled Z gates using the identity:



- Then apply the cosine-sine decomposition inductively.

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- Implement any n -qubit unitary using 1- and 2-qubit gates. *These unitaries can map $|00 \dots 0\rangle$ to any state in $\mathbb{C}^{2^n}$.*

Question: Is this sufficient for quantum advantage? Do the arguments so far show that quantum computers can efficiently perform calculations that classical computers cannot?

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Proof Sketch:

- *Discretize:* any U_n implementable with $\text{poly}(n)$ gates can be approximated to error ε using single-qubit gates accurate to $\varepsilon/\text{poly}(n)$.

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- *Discretize:* any U_n implementable with $\text{poly}(n)$ gates can be approximated to error ε using single-qubit gates accurate to $\varepsilon/\text{poly}(n)$.
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Efficiently Implementable Unitaries

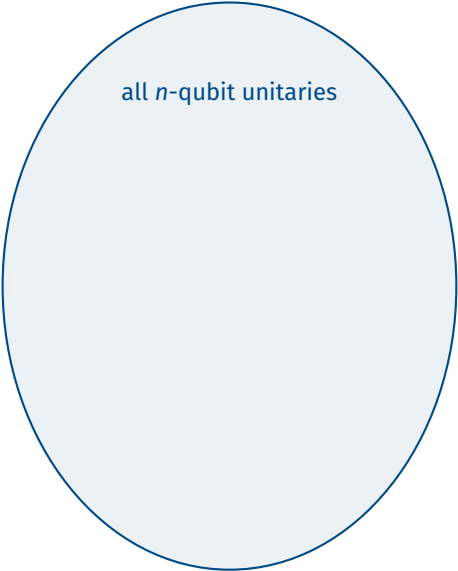
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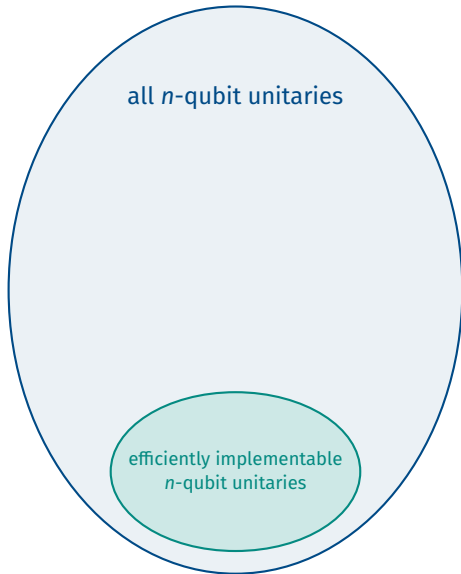
$$\frac{\left((\text{poly}(n)/\varepsilon)^4\right)^{\text{poly}(n)}}{\left(\frac{1}{\varepsilon}\right)^{2^n}} = (\varepsilon)^{2^n} (\text{poly}(n)/\varepsilon)^{4 \text{poly}(n)} \approx (\varepsilon)^{2^n}$$

Efficiently Implementable Unitaries

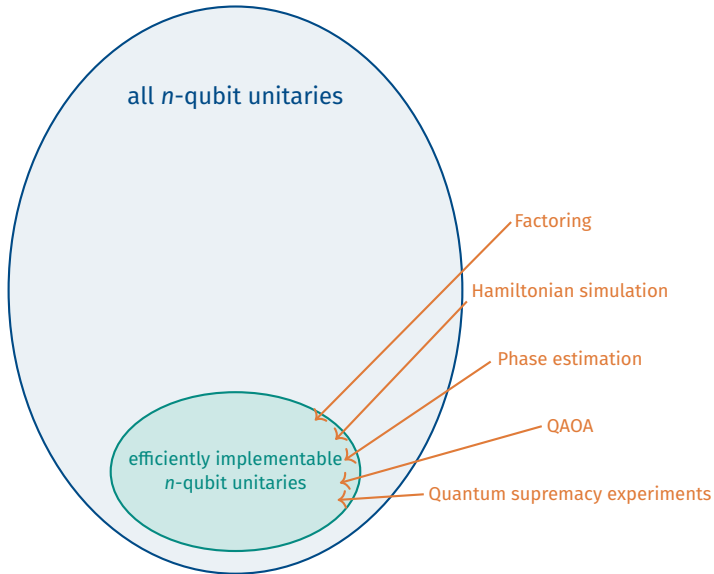


all n -qubit unitaries

Efficiently Implementable Unitaries



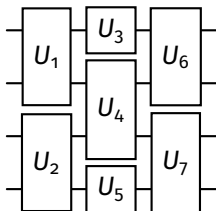
Efficiently Implementable Unitaries



Part 2: Fault-Tolerant Circuits

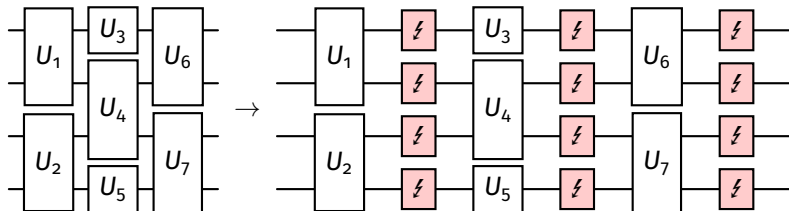
Motivation: Noise

Computations using **physical qubits** accumulate **noise**.



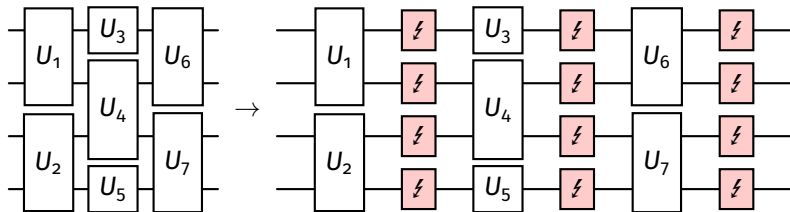
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Implementing complicated unitaries (*Factoring, Phase Estimation, Hamiltonian Simulation*) requires correcting for these errors.

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 - **Easy:** **Pauli updates** $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$.
 - **(Usually) easy:** **Clifford gates** — unitaries that map Pauli operators to Pauli operators under conjugation:

$$\mathcal{C} = \{U : UPU^\dagger \in \mathcal{P}_n \forall P \in \mathcal{P}_n\}$$

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Gottesman-Knill Theorem: Any quantum circuit consisting of:

- Preparation of computational basis states $|x\rangle$
 - Clifford gates
 - Intermediate Pauli measurements and classical feedforward
- can be efficiently simulated on a classical computer.

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- arXiv:2112.02040 (Bouland, Giurgica-Tiron): removes the need for inverses.

Part 3: Quantum Circuit Complexity

Quantum Compilation and Quantum Circuit Complexity

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Quantum Circuit Complexity: What unitaries can be implemented within a given model?

- Computational model might not be practical!
- Constant factors often neglected.
- Interested in quantum advantage and quantum-classical separations.

Sample Circuit Complexity Questions

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- Can shallow quantum circuits perform computation that shallow classical circuits cannot?
 $\rightsquigarrow$ *Unconditional Circuit Separations*
- How hard is it to prepare a target state or implement a target unitary if classical computation is free?
 $\rightsquigarrow$ *State Synthesis, Unitary Synthesis*

References and Further Reading

- **1- and 2-qubit unitaries, measurements**

M. Nielsen and I. Chuang, *Quantum Computation and Quantum Information*, Chapter 4. Cambridge University Press, 2000.

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H. Yuen, *Lecture Notes on Unitary Synthesis*, Spring 2022.

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